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REFLECTIVE LAYERS FOR LIGHT-EMITTING DIODES

Abstract

A light-emitting diode (LED) chip with reflective layers having high reflectivity. The LED chip may include an active LED structure including an active layer between an n-type layer and a p-type layer. A first reflective layer is adjacent the active LED structure and comprises a plurality of dielectric layers with varying optical thicknesses. The plurality of dielectric layers may include a plurality of first dielectric layers and a plurality of second dielectric layers of varying thicknesses and compositions. The LED chip may further include a second reflective layer that includes an electrically conductive path through the first reflective layer.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/336,454, filed Jun. 2, 2021, which is a division of U.S. patent application Ser. No. 15/882,103, filed Jan. 29, 2018, now U.S. Pat. No. 11,031,527, the disclosures of which are hereby incorporated herein by reference in their entirety.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates to solid-state lighting devices including light-emitting diodes and more particularly to light-emitting diodes with reflective layers having high reflectivity.

BACKGROUND

[0003] Solid-state lighting devices such as light-emitting diodes (LEDs) are increasingly used in both consumer and commercial applications. Advancements in LED technology have resulted in highly efficient and mechanically robust light sources with a long service life. Accordingly, modern LEDs have enabled a variety of new display applications and are being increasingly utilized for general illumination applications, often replacing incandescent and fluorescent light sources.

[0004] LEDs are solid-state devices that convert electrical energy to light and generally include one or more active layers of semiconductor material (or an active region) arranged between oppositely doped n-type and p-type layers. When a bias is applied across the doped layers, holes and electrons are injected into the one or more active layers where they recombine to generate emissions such as visible light or ultraviolet emissions. An active region may be fabricated, for example, from silicon carbide, gallium nitride, gallium phosphide, aluminum nitride, and/or gallium arsenide-based materials and/or from organic semiconductor materials. Photons generated by the active region are initiated in all directions.

[0005] Typically, it is desirable to operate LEDs at the highest light emission efficiency, which can be measured by the emission intensity in relation to the output power (e.g., in lumens per watt). A practical goal to enhance emission efficiency is to maximize extraction of light emitted by the active region in the direction of the desired transmission of light. Light extraction and external quantum efficiency of an LED can be limited by a number of factors, including internal reflection. According to the well-understood implications of Snell's law, photons reaching the surface (interface) between an LED surface and the surrounding environment are either refracted or internally reflected. If photons are internally reflected in a repeated manner, then such photons eventually are absorbed and never provide visible light that exits an LED.

[0006] To increase the opportunity for photons to exit an LED, it has been found useful to pattern, roughen, or otherwise texture the interface between an LED surface and the surrounding environment to provide a varying surface that increases the probability of refraction over internal reflection and thus enhances light extraction. Despite the availability of such methods, their practical employment has been limited in at least certain contexts. For example, mechanical methods may introduce stress in or cause breakage of LED material and may also be limited in terms of the position in a fabrication sequence in which they can be employed. Chemical (e.g., photolithographic etching) methods may also be limited in terms of their position in a fabrication sequence to avoid misalignment or microfeature damage during subsequent LED chip fabrication and/or to avoid chemical incompatibility with LED chip layers if etching is performed after chip fabrication.

[0007] Another way to increase light extraction efficiency is to provide reflective surfaces that

reflect generated light so that such light may contribute to useful emission from an LED chip. In a typical LED package **10** illustrated in FIG. **1**, a single LED chip **12** is mounted on a reflective cup **13** by means of a solder bond or conductive epoxy. One or more wire bonds **11** can connect the ohmic contacts of the LED chip **12** to leads **15A** and/or **15B**, which may be attached to or integral with the reflective cup **13**. The reflective cup may be filled with an encapsulant material **16**, which may contain a wavelength conversion material such as a phosphor. At least some of light emitted by the LED at a first wavelength may be absorbed by the phosphor, which may responsively emit light at a second wavelength. The entire assembly is then encapsulated in a clear protective resin **14**, which may be molded in the shape of a lens to collimate the light emitted from the LED chip **12**. While the reflective cup **13** may direct light in an upward direction, optical losses may occur when the light is reflected. Some light may be absorbed by the reflector cup due to the less than 100% reflectivity of practical reflector surfaces. Some metals can have less than 95% reflectivity in the wavelength range of interest.

[0008] FIG. **2** shows another LED package in which one or more LED chips **22** can be mounted onto a carrier such as a printed circuit board (PCB) carrier, substrate, or submount **23**. A metal reflector **24** mounted on the submount surrounds the LED chips **22** and reflects light emitted by the LED chips **22** away from the package **20**. The reflector **24** also provides mechanical protection to the LED chips **22**. One or more wire bond connections **11** are made between ohmic contacts on the LED chips **22** and electrical traces **25A**, **25B** on the submount **23**. The mounted LED chips **22** are then covered with an encapsulant **26**, which may provide environmental and mechanical protection to the chips while also acting as a lens. The metal reflector **24** is typically attached to the carrier by means of a solder or epoxy bond. The metal reflector **24** may also experience optical losses when the light is reflected because it also has less than 100% reflectivity.

[0009] The reflectors shown in FIGS. **1** and **2** are arranged to reflect light that escapes from the LED. LEDs have also been developed that have internal reflective surfaces or layers to reflect light internal to the LEDs. FIG. **3** shows a schematic of an LED chip **30** with an LED **32** mounted on a submount **34** by a metal bond layer **36**. The LED further comprises a p-contact/reflector **38** between the LED **32** and the metal bond **36**, with the p-contact/reflector **38** typically comprising a metal such as silver (Ag). This arrangement is utilized in commercially available LEDs such as those from Cree® Inc., available within the EZBright™ family of LEDs. The p-contact/reflector **38** can reflect light emitted from the LED chip's active region toward the submount back toward the LED's primary emitting surface. The reflector also reflects total internal reflection light back toward the LED's primary emitting surface. Like the metal reflectors above, the p-contact/reflector **38** reflects less than 100% of light and in some cases less than 95%.

[0010] FIG. **4** shows a graph **40** showing the reflectivity of silver on gallium nitride (GaN) at different viewing angles for light with a wavelength of 460 nm. The refractive index of GaN about 2.4, while the complex refractive index for silver is taken from the technical literature. [See *Handbook of Optical Constants of Solids*, edited by E. Palik.] The graph shows the p-polarization reflectivity **42**, s-polarization reflectivity **44**, and average reflectivity **46**, with the average reflectivity **46** generally illustrating the overall reflectivity of the metal for the purpose of LEDs where light is generated with random polarization. The reflectivity at 0 degrees is lower than the reflectivity at 90 degrees, and this difference can result in up to 5% or more of the light being lost on each reflection. In an LED chip, in some instances total internal reflection light can reflect off the mirror several times before it escapes and, as a result, small changes in the mirror absorption can lead to significant changes in the brightness of the LED. The cumulative effect of the mirror absorption on each reflection can reduce the light intensity such that less than 75% of light from the LED's active region actually escapes as LED light.

[0011] The art continues to seek improved light-emitting diodes and solid-state lighting devices having reduced optical losses and providing desirable illumination characteristics capable of overcoming challenges associated with conventional lighting devices.

SUMMARY

[0012] The present disclosure relates to solid-state lighting devices including light-emitting diodes and more particularly to light-emitting diodes with reflective layers having high reflectivity.

[0013] In some embodiments, a light-emitting diode (LED) chip includes an active LED structure comprising an active layer between an n-type layer and a p-type layer and a first reflective layer adjacent the active LED structure and comprising a plurality of dielectric layers. The LED chip includes a second reflective layer on the first reflective layer, a barrier layer on the second reflective layer, and a passivation layer on the barrier layer. In some embodiments, each dielectric layer of the plurality of dielectric layers comprises a different thickness. In some embodiments, the plurality of dielectric layers comprises a plurality of first dielectric layers and a plurality of second dielectric layers. In some embodiments, the plurality of dielectric layers comprises an aperiodic Bragg reflector. In some embodiments, the second reflective layer includes an electrically conductive path through the first dielectric layer.

[0014] In other embodiments, a light-emitting diode (LED) chip includes an active LED structure including an active layer between an n-type layer and a p-type layer, and a first reflective layer adjacent the active LED structure and including a plurality of first dielectric layers and a plurality of second dielectric layers wherein each layer of the plurality of first dielectric layers includes a different thickness. The thickest layer of the plurality of first dielectric layers is between other layers of the plurality of first dielectric layers.

[0015] In other embodiments, a light-emitting diode (LED) chip includes an active LED structure including an active layer between an n-type layer and a p-type layer and a first reflective layer adjacent the active LED structure and including a plurality of first dielectric layers and a plurality of second dielectric layers. An average thickness of the plurality of first dielectric layers is greater than an average thickness of the plurality second dielectric layers and at least one layer of the plurality of second dielectric layers comprises a thickness greater than at least one layer of the plurality of first dielectric layers.

[0016] Those skilled in the art will appreciate the scope of the present disclosure and realize additional aspects thereof after reading the following detailed description of the preferred embodiments in association with the accompanying drawing figures.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

[0017] The accompanying drawing figures incorporated in and forming a part of this specification illustrate several aspects of the disclosure and, together with the description, serve to explain the principles of the disclosure.

[0018] FIG. 1 is a cross-sectional representation of a conventional light-emitting diode (LED).

[0019] FIG. 2 is a cross-sectional representation of a conventional LED.

[0020] FIG. 3 is a partial cross-sectional representation of a conventional LED.

[0021] FIG. 4 is a graph showing the reflectivity of a metal reflector at different viewing angles.

[0022] FIG. 5A is a partial cross-sectional representation of an LED according to some embodiments.

[0023] FIG. 5B is a partial cross-sectional representation of a reflective layer of an LED according to some embodiments.

[0024] FIG. 5C is a partial cross-sectional representation of a reflective layer of an LED according to some embodiments.

[0025] FIG. 5D is a partial cross-sectional representation of a reflective layer of an LED according to some embodiments.

[0026] FIG. 6A is a graph comparing the percent reflectance over a wide wavelength range at a

zero degree angle of incidence for some embodiments.

[0027] FIG. 6B is a heat map representation comparing the reflection intensity for a wide angle of incidence (AOI) range and across a wide wavelength range for some embodiments.

[0028] FIG. 6C is a heat map representation comparing the reflection intensity for a wide angle of incidence (AOI) range and across a wide wavelength range for some embodiments

[0029] FIG. 7 is a graph showing the reflectivity of a reflective layer at different viewing angles according to some embodiments.

[0030] FIG. 8 is a cross-sectional illustration of an LED according to some embodiments.

[0031] FIG. 9 is a microscopic cross-sectional image of a portion of an LED according to some embodiments.

[0032] FIG. 10A is a cross-sectional illustration of an LED according to some embodiments.

[0033] FIG. 10B is a cross-sectional illustration of an LED according to some embodiments.

[0034] FIG. 11A is a cross-sectional illustration of an LED according to some embodiments.

[0035] FIG. 11B is a cross-sectional illustration of an LED according to some embodiments.

[0036] FIG. 12A is a cross-sectional illustration of an LED according to some embodiments.

[0037] FIG. 12B is a cross-sectional illustration of an LED according to some embodiments.

[0038] FIG. 13 is a cross-sectional representation of a packaged LED according to some embodiments.

[0039] FIG. 14 is a cross-sectional representation of a packaged LED according to some embodiments.

[0040] FIG. 15 is a cross-sectional representation of a multiple-junction LED chip according to some embodiments.

[0041] FIG. 16 is a cross-sectional representation of a portion of a lighting fixture according to some embodiments.

DETAILED DESCRIPTION

[0042] The embodiments set forth below represent the necessary information to enable those skilled in the art to practice the embodiments and illustrate the best mode of practicing the embodiments. Upon reading the following description in light of the accompanying drawing figures, those skilled in the art will understand the concepts of the disclosure and will recognize applications of these concepts not particularly addressed herein. It should be understood that these concepts and applications fall within the scope of the disclosure and the accompanying claims.

[0043] It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the present disclosure. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0044] It will be understood that when an element such as a layer, region, or substrate is referred to as being “on” or extending “onto” another element, it can be directly on or extend directly onto the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” or extending “directly onto” another element, there are no intervening elements present. Likewise, it will be understood that when an element such as a layer, region, or substrate is referred to as being “over” or extending “over” another element, it can be directly over or extend directly over the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly over” or extending “directly over” another element, there are no intervening elements present. It will also be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present.

[0045] Relative terms such as “below” or “above” or “upper” or “lower” or “horizontal” or “vertical” may be used herein to describe a relationship of one element, layer, or region to another element, layer, or region as illustrated in the Figures. It will be understood that these terms and those discussed above are intended to encompass different orientations of the device in addition to the orientation depicted in the Figures.

[0046] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes,” and/or “including” when used herein specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0047] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms used herein should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0048] The present disclosure is directed to solid-state emitters having internal or integral reflective surfaces/layers arranged to increase the emission efficiency of the emitters. The present disclosure is described herein with reference to light-emitting diodes (LEDs), but it is understood that it is equally applicable to other solid-state emitters. Devices of the present disclosure can be used as a reflector in conjunction with one or more contacts or can be used as a reflector separate from the contacts.

[0049] As described above, different embodiments of LED chips according to the present disclosure comprise an active LED structure having an active layer between two oppositely doped layers. It is understood that the active LED structure may include many additional layers, and the active layer may include a plurality of layers, and two oppositely doped layers may also include a plurality of layers. A first reflective layer can be provided adjacent to one of the oppositely doped layers, with the first layer comprising a material with a different index of refraction from the active LED structure. In some embodiments, the first reflective layer can comprise a layer with an index of refraction that is primarily lower at or near its interface with the active LED structure. Stated differently, when the layer comprises a number of materials, some embodiments of the layer can have an average index of refraction lower than that of the active LED structure. In still other embodiments, the portion of the reflective layer closest to the active LED structure should be less than that of the active LED structure.

[0050] The difference in index of refraction between the active LED structure and the first reflective layer increases the amount of total internal reflection (TIR) light at this interface. In embodiments in which the first reflective layer has an index of refraction lower than that of the active LED structure, the lower index of refraction material provides a step down in the index of refraction that increases the amount of light that can experience TIR. Some embodiments of LED chips according to the present disclosure can also comprise a second reflective layer or metal layer that can be on and used in conjunction with the first reflective layer such that light passing through the first reflective layer (e.g., not experiencing TIR) can be reflected by the second reflective layer.

[0051] These internal or integral reflective layers can reduce optical emission losses that can occur by light being emitted in an undesirable direction where it can be absorbed. Light that is emitted from the emitter's active LED structure in a direction away from useful light emission, such as toward the substrate, submount, or metal bonding layers, can instead be reflected by the first reflective layer. The reflective surfaces can be positioned to reflect this light so that it emits from the LED chip in a desirable direction. Embodiments of the present disclosure provide one or a

plurality of layers and materials that can cooperate to efficiently reflect light in the desired directions so that it can contribute to the emitter's useful emission.

[0052] The first reflective layer can comprise many different materials including silicon nitride (SiN, SiN_x, Si.sub.3N.sub.4), silicon (Si), germanium (Ge), silicon oxide (SiO.sub.2, SiO_x), titanium oxide (TiO.sub.2), indium tin oxide (ITO), magnesium oxide (MgO_x), zinc oxide (ZnO), and combinations thereof. In some embodiments, the first reflective layer comprises dielectric materials such as silicon dioxide (SiO.sub.2) and/or silicon nitride (SiN, Si.sub.3N.sub.4). It is understood that many other materials can be used with refractive indexes that are lower or higher, with some material having an index of refraction that is up to approximately 50% smaller than that of the LED's active structure material. In other embodiments the index of refraction of the first reflective material can be up to approximately 40% smaller than that of the active structure material, while in other embodiments it can be up to approximately 30% smaller, while in still other embodiments it can be up to approximately 20% smaller. For example, in some embodiments the GaN material of the active structure has an index of refraction of about 2.4, and the reflective material includes one or more layers of silicon dioxide material with an index of refraction of about 1.46 and one or more layers of silicon nitride with an index of refraction of about 1.9.

[0053] Many conventional LEDs rely primarily on a metal reflector layer made of different material such as silver (Ag), Ag alloys, gold (Au), and Au alloys. As described above, there can be losses with each reflection off metal reflectors, and these losses can be significant, particularly for light making multiple passes and reflections in the LED. There are no optical losses in light reflected by TIR, so that when more light is reflected using TIR, the emission efficiency of the LED can increase. Other conventional LEDs chips have relied on internal multiple layer reflectors, such as distributed Bragg reflectors (DBRs). A conventional DBR, such as a quarter-wave reflector, includes multiple pairs of layers with different indexes of refraction. The multiple pairs are arranged sequentially to provide multiple interfaces with index of refraction gradients. Each interface between the two layers contributes a Fresnel reflection; however, this occurs only for a particular angle of incidence range.

[0054] Different embodiments of emitters according to the present disclosure can also utilize other structures, layers, or features that allow for efficient and reliable LED operation. In some embodiments, a current-spreading layer can be included in proximity to the reflective layer to provide for spreading of current into the one or more layers of the active LED structure. In other embodiments, materials can be included to provide for reliable adhesion between different layers, such as between the low index of refraction layer and the metal reflective layer. Different embodiments of the disclosure also provide having conductive via or path arrangements that provide conductive paths through the low index of refraction reflective layer. This allows an electric signal to pass through the low index of refraction layer along the vias so that the composite layer can be used as an internal layer, where an electrical signal can pass through the low index of refraction layer during operation. This via arrangement can take many different shapes and sizes as described in detail below.

[0055] The present disclosure is described herein with reference to certain embodiments, but it is understood that the disclosure can be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. In particular, the lower index of refraction first reflective layer can comprise many different material layers and can have many different thicknesses beyond those described herein. Some embodiments can have layers having layers or materials with an index of refraction higher than that of the active LED structure, but these layers can be thin enough to have minimal optical impact. The first reflective layer can also be in many different locations on different solid-state emitters beyond those described herein and can be used on different devices beyond solid-state emitters. Further, the first reflective layer can be provided with or without conductive structures to allow electrical signals to pass through. It is understood that LEDs according to the present disclosure can also utilize the first reflective layer in

conjunction with other reflectors including metal and other dielectric reflective layers. The first reflective layer is arranged to increase the amount of light reflected by TIR while at the same time maintaining a simple, efficient, and cost-effective reflecting system.

[0056] Embodiments of the disclosure are described herein with reference to cross-sectional view illustrations that are schematic illustrations of embodiments of the disclosure. As such, the actual thickness of the layers can be different, and variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are expected. For example, a region illustrated or described as square or rectangular can have rounded or curved features, and regions shown as straight lines may have some irregularity. Thus, the regions illustrated in the figures are schematic and their shapes are not intended to illustrate the precise shape of a region of a device and are not intended to limit the scope of the disclosure.

[0057] FIG. 5A illustrates a sectional view of an LED chip **50** according to some embodiments. The LED chip **50** includes reflective structures that allow for LED chip operation with increased emission efficiency. Although the present disclosure is described with reference to fabrication of a single LED chip, it is understood that the present disclosure can also be applied to wafer-level LED chip fabrication, fabrication of groups of LED chips, or fabrication of packaged LED chips. The wafer LEDs or groups of LEDs can then be separated into individual LED chips using known singulation or dicing methods. The present disclosure can also be used in different LEDs having different geometries, such lateral geometry or vertical geometry. The present disclosure can also be used in LEDs compatible with flip-chip mounting and in those that are arranged for non-flip-chip mounting.

[0058] The LED chip **50** comprises an active LED structure **52** or region that can have many different semiconductor layers arranged in different ways. The fabrication and operation of LEDs and their active structure are generally known in the art and are only briefly discussed herein. The layers of the active LED structure **52** can be fabricated using known processes with a suitable process being fabrication using metal organic chemical vapor deposition. The layers of the active LED structure **52** can comprise many different layers and generally comprise an active layer sandwiched between n-type and p-type oppositely doped epitaxial layers, all of which are formed successively on a growth substrate. It is understood that additional layers and elements can also be included in the active LED structure **52**, including but not limited to, buffer layers; nucleation layers; super lattice structures; un-doped layers; cladding layers; contact layers; and current-spreading layers and light extraction layers and elements. The active layer can comprise single quantum well, multiple quantum well, double heterostructure, or super lattice structures.

[0059] The active LED structure **52** can be fabricated from different material systems, with some material systems being Group III nitride-based material systems. Group III nitrides refer to those semiconductor compounds formed between nitrogen and the elements in the Group III of the periodic table, usually aluminum (Al), gallium (Ga), and indium (In). Gallium nitride (GaN) is a common binary compound. The term also refers to ternary and quaternary compounds such as aluminum gallium nitride (AlGaN), indium gallium nitride (InGaN), and aluminum indium gallium nitride (AlInGaN). For Group III nitrides, silicon (Si) is a common n-type dopant and magnesium (Mg) is a common p-type dopant. Accordingly, the active layer, n-type layer, and p-type layer may include one or more layers of GaN, AlGaN, InGaN, and AlInGaN that are either undoped or doped with Si or Mg for a material system based on Group III nitrides. Other material systems include silicon carbide, organic semiconductor materials, and other Group III-V systems such as gallium phosphide, gallium arsenide, and related compounds.

[0060] The active LED structure **52** may be grown on a growth substrate (not shown in FIG. 5). Growth substrates can be made of many materials such as sapphire, silicon carbide, aluminum nitride (AlN), GaN, with a suitable substrate being a 4H polytype of silicon carbide, although other silicon carbide polytypes can also be used including 3C, 6H, and 15R polytypes. Silicon carbide has certain advantages, such as a closer crystal lattice match to Group III nitrides than other

substrates and results in Group III nitride films of high quality. Silicon carbide also has a very high thermal conductivity so that the total output power of Group III nitride devices on silicon carbide is not limited by the thermal dissipation of the substrate. Sapphire is another common substrate for Group III nitrides and also has certain advantages, including being lower cost, having established manufacturing processes, and having good light transmissive optical properties.

[0061] Different embodiments of the active LED structure 52 can emit different wavelengths of light depending on the composition of the active layer and n-type and p-type layers. In some embodiments, the active LED structure 52 emits a blue light in the peak wavelength range of approximately 430 nm to 480 nm. In other embodiments, the active LED structure 52 emits green light in a peak wavelength range of 500 nm to 570 nm. In other embodiments, the active LED structure 52 emits red light in a peak wavelength range of 600 to 650 nm. The LED chip 50 can also be covered with one or more lumiphors or other conversion materials, such as phosphors, such that at least some of the light from the LED passes through the one or more phosphors and is converted to one or more different wavelengths of light. In one embodiment, the LED chip emits a white light combination of light from the LED's active structure and light from the one or more phosphors. The one or more phosphors may include yellow (e.g., YAG:Ce), green (LuAg:Ce), and red (Ca.sub.i-x-ySr.sub.xEu.sub.yAlSiN.sub.3) emitting phosphors, and combinations thereof.

[0062] As mentioned above, different layers can be included to allow for efficient operation of the LED chip 50. For some semiconductor materials, such as some Group III nitride material systems, current does not effectively spread through the p-type layer. In these embodiments, a current-spreading layer 54 can be on the active LED structure 52 in a location to aid in current-spreading into the p-type layer. In some embodiments the current-spreading layer 54 can cover some or the entire p-type layer, and in some embodiments the current-spreading layer 54 helps spread current from a p-type contact across the surface of the p-type layer as described in more detail below. This helps to provide improved current-spreading across the p-type layer with a corresponding improvement in current injection from the p-type layer.

[0063] The current-spreading layer 54 can comprise many different materials and is typically a transparent conductive oxide such as ITO or a metal such as platinum (Pt), although other materials can also be used. The current-spreading layer can have many different thicknesses, with the present disclosure having a thickness small enough to minimize absorption of light from the active structure that passes through the current-spreading layer. Some embodiments of current-spreading layer 54 comprising ITO can have thicknesses less than 1000 angstroms (Å), while other embodiments can have a thickness less than 700 Å. Still other embodiments can have a thickness less than 500 Å. Still other embodiments can have a thickness in the range of 50 Å to 300 Å, with some of these embodiments having a current-spreading layer with a thickness of approximately 200 Å. The current-spreading layer 54 and the reflective layers described below can be deposited using known methods. It is understood that in embodiments where current spreading is not a concern, the LED chips can be provided without a current-spreading layer.

[0064] The LED chip 50 can also comprise a first reflective layer 56 that can be formed on the active LED structure 52, and in the embodiment shown is formed on the current-spreading layer 54 with current-spreading layer 54 between a first reflective layer 56 and the active LED structure 52. The first reflective layer 56 can comprise many different materials and preferably comprises a material that presents an index of refraction step with the material comprising the active LED structure 52. Stated differently, the first reflective layer 56 should have an index of refraction that is smaller than that of the active LED structure 52 to promote TIR of active structure light-emitting toward the first reflective layer 56, as shown by first light trace 58. Light that experiences TIR is reflected without experiencing absorption or loss, and TIR allows for the efficient reflection of active structure light so that it can contribute to useful or desired LED chip emission. In embodiments that rely solely on metal layers to reflect light, the light experiences loss with each reflection (as described above), which can reduce overall LED chip emission efficiency.

[0065] In some embodiments, the first reflective layer **56** comprises a material with an index of refraction lower than the index of refraction of the active LED structure **52** material, with the lower index of refraction material providing for increased TIR of light from the active LED structure **52**. The first reflective layer **56** may comprise many different materials, with some having an index of refraction less than 2.3, while others can have an index of refraction less than 2.15, less than 2.0, and less than 1.5. In some embodiments the first reflective layer **56** comprises a dielectric material, with some embodiments comprising silicon dioxide and/or silicon nitride. It is understood that many dielectric materials can be used such as SiN, SiNx, Si.sub.3N.sub.4, Si, Ge, SiO.sub.2, SiOx, TiO.sub.2, Ta.sub.2O.sub.5, ITO, MgOx, ZnO, and combinations thereof.

[0066] As mentioned above, some Group III nitride materials such as GaN can have an index of refraction of approximately 2.4, and silicon dioxide can have an index of refraction of approximately 1.48, and silicon nitride can have an index of refraction of approximately 1.9. Embodiments with an active LED structure **52** comprising GaN and the first reflective layer **56** that comprises silicon dioxide can have a sufficient index of refraction step between the two to allow for efficient TIR of light at the junction between the two as shown by first light trace **58**. The first reflective layer **56** can have different thicknesses depending on the type of materials used, with some embodiments having a thickness of at least 0.2 microns (μm). In some of these embodiments it can have a thickness in the range of 0.2 μm to 0.7 μm , while in some of these embodiments it can be approximately 0.5 μm thick.

[0067] As light experiences TIR at the junction with the first reflective layer **56**, an evanescent wave with exponentially decaying intensity can extend into the first reflective layer **56**. This wave is most intense within approximately one third of the light wavelength from the junction (about 0.3 μm for 450 nm light in silicon dioxide). If the thickness of the first reflective layer **56** is too thin, such that significant intensity remains in the evanescent wave at the interface between the first reflective layer **56** and a second reflective layer **60**, a portion of the light can reach the second reflective layer **60**. This in turn can reduce the TIR reflection at the first interface. For this reason, in some embodiments the first reflective layer **56** should have a thickness of at least 0.3 μm .

[0068] As mentioned above, an LED chip **50** according to some embodiments can also utilize the second reflective layer **60**, such as a metal layer, on the first reflective layer **56** to reflect any light that may pass through the first reflective layer **56**. An example of the light reflected at the second reflective layer **60** is shown by second light trace **62**, which passes first through the first reflective layer **56** before being reflected at the second reflective layer **60**. The second reflective layer **60** can comprise many different materials such as Ag, Au, Al, or combinations thereof, with the present embodiment being Ag. Some embodiments may also comprise an adhesion layer **64** between the first reflective layer **56** and the second reflective layer **60** to promote adhesion between the two. Many different materials can be used for the adhesion layer **64**, such as TiO, TiON, TiO.sub.2, tantalum oxide (TaO), TaON, Ta.sub.2O.sub.5, AlO, or combinations thereof, with a preferred material being TiON. The adhesion layer **64** may have many different thicknesses from just a few angstroms to thousands of angstroms. In some embodiments it can be less than 100 Å, while in other embodiments it can be less than 50 Å. In some of these embodiments, it can be approximately 20 Å thick. The thickness of the adhesion layer **64** and the material used should minimize the absorption of light passing to minimize losses of light reflecting off the second reflective layer **60**.

[0069] As stated above, the interface between the first reflective layer **56** and the active LED structure **52** includes an index of refraction step to promote TIR of light, as shown by first light trace **58**. However, depending on the angle of incidence, some light may still pass through the first reflective layer **56** before being reflected at the second reflective layer **60**, as depicted by the second light trace **62**. FIG. 5B illustrates a sectional illustration of the first reflective layer **56** according to some embodiments. The first reflective layer **56** includes a plurality of layers (**56-1**, **56-2**, **56-3**, **56-4**, **56-5**, **56-6**, and **56-7**) configured to provide a plurality of different interfaces between them. Additionally, layer **56-1** forms an interface with the active LED structure **52** and

layer 56-7 forms an interface with either the adhesion layer 64 or the second reflective layer 60 of FIG. 5A. Each different interface promotes TIR of light having a different angle of incidence range and accordingly, the total amount of light reflected by the first reflective layer 56 is increased, and the amount of light reaching the second reflective layer 60 is reduced.

[0070] A different interface may be formed from materials having different indexes of refraction. For example, layer 56-1 may include silicon dioxide and layer 56-2 may include silicon nitride. Accordingly, the interface between layer 56-1 and the active LED structure 52 is different from the interface between layer 56-1 and layer 56-2. If two interfaces are provided between the same two materials, then the interfaces may be different by varying the optical thickness of each layer. Optical thickness can be defined as the product of the refractive index of the material and the geometric length the path of light travels. Accordingly, the optical thickness of a layer of material may be changed by increasing or decreasing the actual layer thickness. A layer with a larger optical thickness will generally promote TIR of light having shallower angles of incidence than another layer with a smaller optical thickness. Accordingly, a plurality of layers with varying optical thicknesses allow some layers to reflect more light of shallower angles of incidence while having other layers that reflect more light at greater angles of incidence, thus providing the plurality of layers with increased total reflection over all angles.

[0071] In some embodiments, the first reflective layer 56 includes a plurality of first dielectric layers 56-1, 56-3, 56-5, and 56-7 of a first material and a plurality of second dielectric layers 56-2, 56-4, and 56-6 of a second material that is different from the first material. If the thickness of each of the plurality of first dielectric layers 56-1, 56-3, 56-5, and 56-7 is varied, then each of the interfaces, that is, 56-1 and 56-2, 56-2 and 56-3, 56-3 and 56-4, 56-4 and 56-5, and 56-5 and 56-6, will be different, and accordingly, each interface may promote TIR of light having different angle of incidence ranges. In some embodiments, the thickness of each of the plurality of second dielectric layers 56-2, 56-4, and 56-6 may also be varied. In some embodiments, the first material is silicon dioxide and the second material is silicon nitride. In other embodiments, the first and second material may be any combination of SiN, SiNx, Si.sub.3N.sub.4, Si, Ge, SiO.sub.2, SiOx, TiO.sub.2, Ta.sub.2O.sub.5, ITO, MgOx, ZnO, or related materials. As illustrated in FIG. 5B, some embodiments include an uneven number of first dielectric layers and an even number of second dielectric layers. In other embodiments, the number of the first dielectric layers is equal to the number of the second dielectric layers. The thickness of the plurality of first dielectric layers 56-1, 56-3, 56-5, and 56-7 may be varied in different configurations within the first reflective layer 56. For example, the thickness of each layer of the plurality of first dielectric layers 56-1, 56-3, 56-5, and 56-7 may increase or decrease sequentially within the first reflective layer 56. In other embodiments, a thickest layer (illustrated as layer 56-3 in FIG. 5B) of the plurality of first dielectric layers 56-1, 56-3, 56-5, and 56-7 is between other layers (56-1, 56-5, and 56-7) of the plurality of first dielectric layers 56-1, 56-3, 56-5, and 56-7. The thickest layer (layer 56-3 in FIG. 5B) has the longest optical thickness and is good for promoting TIR of light having the shallowest angles of incidence, such as 0 to 15 degrees, compared with the other layers (layers 56-1, 56-5, and 56-7). By placing the thickest layer (layer 56-3) between the other layers (layers 56-1, 56-5, and 56-7), at least some light with greater angles of incidence, such as greater than 15 degrees, may be reflected earlier without potentially being lost to absorption within the first reflective layer 56. In FIG. 5B, the thickest layer of the plurality of first dielectric layers 56-1, 56-3, 56-5, and 56-7 is illustrated as layer 56-3; however, layer 56-5 could also be the thickest layer without deviating from the principles of these embodiments.

[0072] Accordingly, some embodiments disclosed herein include an LED chip that includes a first reflective layer that includes a plurality of dielectric layers with varying optical thicknesses. In some embodiments, an LED chip comprising an active structure comprising an active layer between an n-type layer and a p-type layer; a first reflective layer adjacent the active LED structure and comprising a plurality of first dielectric layers and a plurality of second dielectric layers

wherein each layer of the plurality of first dielectric layers comprises a different thickness; wherein a thickest layer of the plurality of first dielectric layers is between other layers of the plurality of first dielectric layers.

[0073] In some embodiments, an average thickness of the plurality of first dielectric layers **56-1**, **56-3**, **56-5**, and **56-7** is greater than an average thickness of the plurality of second dielectric layers **56-2**, **56-4**, and **56-6**. However, in some embodiments, at least one layer of the plurality of second dielectric layers **56-2**, **56-4**, and **56-6** has a thickness greater than at least one of the plurality of first dielectric layers **56-1**, **56-3**, **56-5**, and **56-7**. For example, the plurality of first dielectric layers **56-1**, **56-3**, **56-5**, and **56-7** may comprise silicon dioxide, and the plurality of second dielectric layers **56-2**, **56-4**, and **56-6** comprise silicon nitride, and the thickness of layer **56-1** is from 190 nm to 200 nm, the thickness of layer **56-2** is from 50 nm to 60 nm, the thickness of layer **56-3** is from 335 nm to 345 nm, the thickness of layer **56-4** is from 55 nm to 65 nm, the thickness of layer **56-5** is from 75 nm to 85 nm, the thickness of layer **56-6** is from 60 nm to 70 nm, and the thickness of layer **56-7** is from 45 nm to 55 nm. In some embodiments, the average thickness of the plurality of first dielectric layers is greater than the average thickness of the plurality of second dielectric layers by at least a factor of 2. In other embodiments, the average thickness of the plurality of first dielectric layers is greater than the average thickness of the plurality of second dielectric layers by at least a factor of 3.

[0074] FIG. 5C illustrates a sectional illustration of the first reflective layer **56** according to other embodiments. In FIG. 50, the first reflective layer **56** includes a plurality of layers (**56-1**, **56-2**, **56-3**, **56-4**, **56-5**, **56-6**, **56-7**, **56-8**, **56-9**, **56-10**, **56-11**, **56-12**, **56-13**) configured to provide a plurality of different interfaces between them. Layer **56-1** forms an interface with the active LED structure **52** of FIG. 5A and layer **56-7** forms an interface with either the adhesion layer **64** or the second reflective layer **60** of FIG. 5A. Each different interface promotes TIR of light having a different angle of incidence range, and accordingly, the total amount of light reflected by the first reflective layer **56** is increased and the amount of light reaching the second reflective layer **60** is reduced. In some embodiments, the first reflective layer **56** includes a plurality of first dielectric layers **56-1**, **56-3**, **56-5**, **56-7**, **56-9**, **56-11**, and **56-13** of a first material and a plurality of second dielectric layers **56-2**, **56-4**, **56-6**, **56-8**, **56-10**, and **56-12** of a second material that is different from the first material. The first material and the second material may be any material or any combination of the materials described above for FIG. 5B. The thicknesses of each layer of the plurality first dielectric layers and each layer of the plurality of second dielectric layers may be varied as described above for FIG. 5B.

[0075] In FIG. 5C, the thickest layer is illustrated as layer **56-1**, which forms an interface with the active LED structure of FIG. 5A. As described above for FIG. 5B, the thickest layer (layer **56-1** in FIG. 5C) has the longest optical thickness and is good for promoting TIR of light having the shallowest angles of incidence, such as 0 to 15 degrees, compared with the other layers. In order to reduce the chances that light of greater angles of incidence, such as greater than 15 degrees, is lost to absorption, additional layers (layers **56-8**, **56-9**, **56-10**, **56-11**, **56-12**, and **56-13** when compared with FIG. 5B) with different optical thicknesses are provided. In some embodiments, the thickest layer (layer **56-1**) is at least 5 times thicker than any other layer of the plurality of first dielectric layers (layers **56-3**, **56-5**, **56-7**, **56-9**, **56-11**, and **56-13**) and the plurality of second dielectric layers (layers **56-2**, **56-4**, **56-6**, **56-8**, **56-10**, and **56-12**). In some embodiments, the thinnest layer (layer **56-13**) of the plurality of first dielectric layers forms an interface with either the adhesion layer **64** or the second reflective layer **60** of FIG. 5A.

[0076] In some embodiments, an average thickness of the plurality of first dielectric layers (layers **56-1**, **56-3**, **56-5**, **56-7**, **56-9**, **56-11**, and **56-13**) is at least 2 times greater than an average thickness of the plurality of second dielectric layers (layers **56-2**, **56-4**, **56-6**, **56-8**, **56-10**, and **56-12**). In further embodiments, at least one layer of the plurality of second dielectric layers (layers **56-2**, **56-4**, **56-6**, **56-8**, **56-10**, and **56-12**) is at least 2 times thicker than the thinnest layer (**56-13**) of the

plurality of first dielectric layers (layers **56-1**, **56-3**, **56-5**, **56-7**, **56-9**, **56-11**, and **56-13**). In some embodiments, the plurality of first dielectric layers (layers **56-1**, **56-3**, **56-5**, **56-7**, **56-9**, **56-11**, and **56-13**) comprises silicon dioxide, and the plurality of second dielectric layers (layers **56-2**, **56-4**, **56-6**, **56-8**, **56-10**, and **56-12**) comprises silicon nitride, and the thickness of layer **56-1** is from 480 nm to 490 nm, the thickness of layer **56-2** is from 80 nm to 90 nm, the thickness of layer **56-3** is from 50 nm to 60 nm, the thickness of layer **56-4** is from 55 nm to 65 nm, the thickness of layer **56-5** is from 65 nm to 75 nm, the thickness of layer **56-6** is from 60 nm to 70 nm, the thickness of layer **56-7** is from 65 nm to 75 nm, the thickness of layer **56-8** is from 55 nm to 65 nm, the thickness of layer **56-9** is from 50 nm to 60 nm, the thickness of layer **56-10** is from 70 nm to 80 nm, the thickness of layer **56-11** is from 80 nm to 90 nm, the thickness of layer **56-12** is from 65 nm to 75 nm, and the thickness of layer **56-13** is from 45 nm to 55 nm.

[0077] FIG. 5D illustrates a sectional illustration of the first reflective layer **56** according to other embodiments. The embodiments of FIG. 5D are similar to the embodiments of FIG. 5C.

Accordingly, the description of FIG. 5C also applies to FIG. 5D with differences provided below.

[0078] In FIG. 5D, the thickest layer is also illustrated as layer **56-1**, which forms an interface with the active LED structure **52** of FIG. 5A. The second thickest layer is illustrated as layer **56-3** and both layers **56-1** and **56-3** are notably thicker than any other layer of the first reflective layer **56**.

Accordingly, layers **56-1** and **56-3** have the largest two optical thicknesses and promote TIR of light having shallower angles of incidence than the other layers. Layer **56-3** has a different optical thickness than layer **56-1** and will trade off some shallow angle reflection to increase reflection at greater angles, thus helping increase the total reflection over all angles. In some embodiments, the thickest layer (layer **56-1**) is at least 2 times thicker than the second thickest layer (layer **56-3**) and at least 5 times thicker than any other layer of the plurality of first dielectric layers (layers **56-5**, **56-7**, **56-9**, **56-11**, and **56-13**). Additionally, the thickest layer (layer **56-1**) is at least 6 times thicker than any of the plurality of second dielectric layers (layers **56-2**, **56-4**, **56-6**, **56-8**, **56-10**, and **56-12**). When the first reflective layer **56** of FIG. 5A is configured as illustrated in FIG. 5D, layers **56-1** and **56-3** are closer to the active LED structure **52** than any other layer of the plurality of first dielectric layers (**56-5**, **56-7**, **56-9**, **56-11**, and **56-13**). Accordingly, more light with shallow angles of incidence are reflected earlier than the embodiments described for FIG. 5C. In turn, at least one other layer of the plurality of first dielectric layers (layers **56-5**, **56-7**, **56-9**, **56-11**, and **56-13**) is much thinner than any other layer. In FIG. 5D, the thinnest layer is illustrated as layer **56-7**, but it is understood in various embodiments that the thinnest layer could be any of layers (layers **56-5**, **56-7**, **56-9**, **56-11**, and **56-13**). In some embodiments, the thickest layer (layer **56-1**) is at least 10 times thicker than the thinnest layer (layer **56-7**) of the plurality of first dielectric layers and in further embodiments, the thickest layer (layer **56-1**) is at least 16 times thicker than the thinnest layer (layer **56-7**). Additionally, the thickest layer (layer **56-4**) of the plurality of second dielectric layers is at least 2 times thicker than the thinnest layer (layer **56-7**) of the plurality of first dielectric layers, and the thinnest layer (layer **56-6**) of the plurality of second dielectric layers is at least one and a half times thicker than the thinnest layer (layer **56-7**).

[0079] In some embodiments, the plurality of first dielectric layers (layers **56-1**, **56-3**, **56-5**, **56-7**, **56-9**, **56-11**, and **56-13**) comprises silicon dioxide, and the plurality of second dielectric layers (layers **56-2**, **56-4**, **56-6**, **56-8**, **56-10**, and **56-12**) comprises silicon nitride, and the thickness of layer **56-1** is from 490 nm to 510 nm, the thickness of layer **56-2** is from 60 nm to 70 nm, the thickness of layer **56-3** is from 195 nm to 205 nm, the thickness of layer **56-4** is from 70 nm to 80 nm, the thickness of layer **56-5** is from 90 nm to 100 nm, the thickness of layer **56-6** is from 40 nm to 50 nm, the thickness of layer **56-7** is from 25 nm to 35 nm, the thickness of layer **56-8** is from 70 nm to 80 nm, the thickness of layer **56-9** is from 70 nm to 80 nm, the thickness of layer **56-10** is from 65 nm to 75 nm, the thickness of layer **56-11** is from 80 nm to 90 nm, the thickness of layer **56-12** is from 60 nm to 70 nm, and the thickness of layer **56-13** is from 55 nm to 65 nm.

[0080] Accordingly, the first reflective layer **56** of FIG. 5A may comprise a plurality of dielectric

layers. The plurality of dielectric layers may comprise embodiments in which each dielectric layer of the plurality of dielectric layers comprises a different thickness. In some embodiments, the first reflective layer **56** comprises from 7 to 13 dielectric layers. In some embodiments, the first reflective layer **56** comprises a plurality of alternating first dielectric layers and second dielectric layers. The first dielectric layers may have a different index of refraction than the second dielectric layers. In some embodiments, the plurality of first dielectric layers comprises silicon dioxide and the plurality of second dielectric layers comprises silicon nitride, although other material combinations are possible as described above. In some embodiments, the first reflective layer **56** comprises an uneven number of first dielectric layers and second dielectric layers, while in other embodiments, the first reflective layer **56** comprises an even number of first dielectric layers and of second dielectric layers. In some embodiments, the first reflective layer **56** comprises an aperiodic Bragg reflector. In some embodiments, the first reflective layer **56** comprises a plurality of dielectric layers in which the thickest dielectric layer of the plurality of dielectric layers is spaced from the active layer by at least one thinner dielectric layer. In other embodiments, the thickest dielectric layer is adjacent the active LED structure and in further embodiments, the second thickest dielectric layer is adjacent the thickest dielectric layer when compared with other layers of the plurality of first dielectric layers. In some embodiments, the first reflective layer **56** comprises a plurality of first dielectric layers having an average thickness that is greater than an average thickness of a plurality of second dielectric layers, and at least one layer of the plurality of second dielectric layers has a thickness greater than at least one layer of the plurality of first dielectric layers. In some embodiments, the thickest layer of the plurality of first dielectric layers is at least 10 times thicker than the thinnest layer of the plurality of first dielectric layers, and in further embodiments, the thickest layer of the plurality of first dielectric layers is at least 16 times thicker than the thinnest layer of the plurality of first dielectric layers.

[0081] Accordingly, some embodiments disclosed herein include an LED chip comprising an active LED structure comprising an active layer between an n-type layer and a p-type layer, and a first reflective layer adjacent the active LED structure and comprising a plurality of first dielectric layers and a plurality of second dielectric layers, wherein an average thickness of the plurality of first dielectric layers is greater than an average thickness of the plurality second dielectric layers, and wherein at least one layer of the plurality of second dielectric layers comprises a thickness greater than at least one layer of the plurality of first dielectric layers.

[0082] As discussed above, in some embodiments, the first reflective layer **56** comprises a plurality of alternating first and second dielectric layers of varying materials with different indexes of refraction and thicknesses that promote TIR over a wide range of angles of incidence. Accordingly, the first reflective layer **56** may be referred to as an aperiodic Bragg reflector. In contrast, a periodic Bragg reflector typically consists of multiple pairs of two materials with alternating higher and lower indexes of refraction. The thickness of each material is chosen so that reflected waves are in constructive interference, and the thickness of each material is kept constant among the repeating pairs. Periodic Bragg reflectors offer improved reflectivity over single dielectric reflectors or metal reflectors, but the improved reflectivity is limited to a particular angle of incidence range.

[0083] FIG. **6A** is a graph comparing the percent reflectance over a wide wavelength range at a zero degree angle of incidence for some embodiments. Each embodiment includes a first reflective layer and a second reflective layer on an active LED structure. Line **66** represents an embodiment in which the first reflective layer is a single dielectric layer of silicon dioxide and the second reflective layer is silver. Line **68** represents an embodiment in which the first reflective layer comprises a plurality of first and second dielectric layers including silicon dioxide and silicon nitride with compositions and thicknesses as described above, and the second reflective layer is silver. Line **68** has noticeably higher reflectivity at most wavelengths. For example, the reflectivity is at least 2% higher at some wavelengths above 450 nm and about 5% higher at some wavelengths

from 500 nm to 530 nm.

[0084] FIG. 6B and FIG. 6C are heat map representations comparing reflection intensity across a wide angle of incidence (AOI) range and across a wide wavelength range. As with FIG. 6A, the embodiments represented by FIG. 6B and FIG. 6C both include a first reflective layer and a second reflective layer on an active LED structure. FIG. 6B represents an embodiment in which the first reflective layer comprises a plurality of first and second dielectric layers including silicon dioxide and silicon nitride with compositions and thicknesses as described above, and the second reflective layer is silver, as for example, the LED chip of FIG. 5A with the first reflective layer 56 of FIG. 5D. FIG. 6C represents an embodiment in which the first reflective layer is a single dielectric layer of silicon dioxide and the second reflective layer is silver. Notably, the percent reflectance improvements illustrated in FIG. 6A for zero angle of incidence are also realized across a wide angle of incidence range. For example, at a most wavelengths from 400 nm to 650 nm, the embodiment of FIG. 6B has greater reflection intensity (represented as darker regions) than the embodiment of FIG. 6C across the most angles of incidence.

[0085] FIG. 7 is a graph representing the reflectivity percent (%) of a sample including a first reflective layer and a second reflective layer on an active LED structure. The first reflective layer comprises a plurality of first and second dielectric layers including silicon dioxide and silicon nitride with compositions and thicknesses as described above, and the second reflective layer is silver, as for example, the LED chip of FIG. 5A with the first reflective layer 56 of FIG. 5D. The reflectivity percent (y-axis) is plotted at different angles of incidence (x-axis) for light with a wavelength of 460 nm. The graph shows the p-polarization reflectivity 70, s-polarization reflectivity 72, and average reflectivity 74, with the average reflectivity 74 generally illustrating the overall reflectivity of the first reflective layer for the purpose of LEDs in which light is generated with random polarization. Notably, the reflectivity is substantially improved when compared with the similar graph of FIG. 4 that represents only a silver reflective layer on GaN. For example, the average reflectivity 74 at zero degrees is at least 93% and exceeds 94% for most angles from about 15 degrees to over 50 degrees.

[0086] It is understood that the first reflective layer 56 arrangements described above can be used in many different LED chips according to the present disclosure. FIG. 8 illustrates some embodiments of an LED chip 80 having a lateral geometry and arranged for flip-chip mounting. The LED chip 80 comprises an active LED structure 82 comprising a p-type layer 84, n-type layer 86, and an active layer 88 formed on a substrate 91. In some embodiments, the n-type layer 86 is between the active layer 88 and the substrate 91. In other embodiments, the p-type layer 84 is between the active layer 88 and the substrate 91. The substrate 91 can comprise many different materials such as silicon carbide or sapphire and can have one or more surfaces that are shaped, textured, or patterned to enhance light extraction.

[0087] The LED chip 80 also comprises a current-spreading layer 90 that is between the active LED structure 82 and a first reflective layer 92. The current-spreading layer 90 can have the same thickness and can comprise the same materials as the current-spreading layer 54 shown in FIG. 5A and described above. In LED chip 80, the current-spreading layer 90 can comprise ITO and is on the p-type layer 84 to spread current into the p-type layer 84. The first reflective layer 92 is arranged on the current-spreading layer 90 and adjacent the p-type layer 84 and can have any of the embodiments with a plurality of layers previously described for the first reflective layer 56, for example, as described for FIG. 5B. For example, in LED chip 80 the first reflective layer 92 may comprise a plurality of dielectric layers. The plurality of dielectric layers may comprise embodiments in which each dielectric layer of the plurality of dielectric layers comprises a different thickness. In some embodiments, the first reflective layer 92 comprises from 7 to 13 dielectric layers. In some embodiments, the first reflective layer 92 comprises a plurality of alternating first dielectric layers and a plurality of alternating second dielectric layers. The plurality of first dielectric layers may have a different index of refraction than the plurality second dielectric

layers. In further embodiments, the plurality of first dielectric layers comprises silicon dioxide and the plurality of second dielectric layers comprises silicon nitride, although other material combinations are possible as described above. In some embodiments, the first reflective layer **92** comprises an uneven number of first dielectric layers and second dielectric layers, while in other embodiments, the first reflective layer **92** comprises an even number of first dielectric layers and of second dielectric layers. In some embodiments, the first reflective layer **92** comprises an aperiodic Bragg reflector. In some embodiments, the first reflective layer **92** comprises a plurality of dielectric layers in which the thickest dielectric layer of the plurality of dielectric layers is spaced from the active layer by at least one thinner dielectric layer. In some embodiments, the first reflective layer **92** comprises a plurality of first dielectric layers having an average thickness that is greater than an average thickness of a plurality of second dielectric layers, and at least one layer of the plurality of second dielectric layers has a thickness greater than at least one layer of the plurality of first dielectric layers. All other embodiments described for the first reflective layer **56** (FIG. 5B) are also applicable to the first reflective layer **92**.

[0088] A second reflective layer **94** and an adhesion layer **96** are included on the first reflective layer **92**, with the adhesion layer **96** sandwiched between and providing adhesion between the second reflective layer **94** and first reflective layer **92**. These layers can comprise the same material and can have the same thickness as the second reflective layer **60** and adhesion layer **64** described above for FIG. 5A. For example, in some embodiments, the second reflective layer **94** comprises an electrically conductive material, such as silver or other metals.

[0089] The LED chip **80** further comprises reflective layer holes **98** that can pass through the adhesion layer **96** and the first reflective layer **92** to the current-spreading layer **90**. The holes **98** can then be filled or partially filled when the second reflective layer **94** is deposited. Accordingly, the second reflective layer **94** is formed on the first reflective layer **92** and comprises vias **100** to the current-spreading layer **90**. As described in more detail below, second reflective layer **94**, by way of vias **100**, provides an electrically conductive path through the first reflective layer **92**, between a p-contact **110** and the current-spreading layer **90**. In some embodiments, the second reflective layer **94** completely fills the holes **98**. In other embodiments, the second reflective layer **94** only partially fills the holes **98**. When the second reflective layer **94** only partially fills the holes **98**, then a barrier layer **102** and a passivation layer **106** may fill the remaining portion of the vias.

[0090] The holes **98** can be formed using many known processes such as conventional etching processes or mechanical processes such as microdrilling.

[0091] The holes **98** can have many different shapes and sizes, with the holes **98** in the embodiment shown having angled or curved side surfaces and a circular cross-section with a diameter of less than 20 μm . In some embodiments, the holes **98** can have a diameter of approximately 8 μm , with others having a diameter down to 1 μm . Adjacent holes **98** can be less than 100 μm apart, with the embodiment shown having a spacing of 30 μm from edge to edge. In still other embodiments, the holes **98** can have a spacing of as small as 10 μm or less. It is understood that the holes **98** (and resulting vias **100**) can have cross-section with different shapes such as square, rectangular, oval, hexagon, and pentagon. In other embodiments the holes are not uniform size and shapes, and there can be different or non-uniform spaces between adjacent holes.

[0092] In other embodiments, different structures can be used to provide a conductive path between the p-contact **110** and the current-spreading layer **90**. Instead of holes **98**, an interconnected grid can be formed through the first reflective layer **92**, with a conductive material then being deposited in the grid to form the conductive path to the current-spreading layer **90**. The grid can take many different forms, with portions of the grid interconnecting at different angles in different embodiments. An electrical signal applied to the grid can spread throughout and along the interconnected portions. It is further understood that in different embodiments a grid can be used in combination with holes, while other embodiments can provide other conductive paths. In some embodiments one or more conductive paths can run outside the LED chip's active layer, such as

along a side surface of the LED chip.

[0093] The LED chip **80** can also comprise a barrier layer **102** on the second reflective layer **94** to prevent migration of the second reflective layer **94** material, such as Ag, to other layers. Preventing this migration helps the LED chip **80** maintain efficient operation through its lifetime. Accordingly, the barrier layer **102** is also part of the conductive path from the p-contact **110** to the current-spreading layer **90**. In some embodiments, the barrier layer **102** is a single layer, and in other embodiments, the barrier layer **102** comprises a plurality of layers. Suitable materials for the barrier layer **102** include but are not limited to sputtered Ti/Pt followed by evaporated Au bulk material or sputtered Ti/Ni followed by a evaporated Ti/Au bulk material.

[0094] An active structure hole **104** can be included passing through the adhesion layer **96**, the first reflective layer **92**, and p-type layer **84** to expose the n-type layer **86**. A passivation layer **106** is included on the barrier layer **102** and the side surfaces of the active structure hole **104**. The passivation layer **106** protects and provides electrical insulation between the contacts and the layers below as described in more detail below. The passivation layer **106** can comprise many different materials, such as a dielectric material. In some embodiments, the passivation **106** is a single layer, and in other embodiments, the passivation layer **106** comprises a plurality of layers. A suitable material for the passivation layer **106** includes but is not limited to silicon nitride.

[0095] Passivation layer hole **108** can be formed through the passivation layer **106** to the barrier layer **102** and/or the second reflective layer **94**. The p-contact **110** can then be deposited in the passivation layer hole **108**. In operation, an electrical signal applied to the p-contact passes through the barrier layer **102**, through the second reflective layer **94** and the vias **100**, and to the current-spreading layer **90** through which it is spread to the p-type layer **84**. Similarly, an n-contact **112** is formed on the passivation layer **106** and through the active structure hole **104**, with the n-contact **112** providing an electrical path for an electrical signal to be applied to the n-type layer **86**. In operation, a signal applied across the p-contact **110** and the n-contact **112** is conducted to the p-type layer **84** and the n-type layer **86**, causing the LED chip **80** to emit light from its active layer **88**.

[0096] The p-contact **110** and the n-contact **112** can comprise many different materials such as Au, copper (Cu), nickel (Ni), In, Al, Ag, tin (Sn), Pt, or combinations thereof. In still other embodiments they can comprise conducting oxides and transparent conducting oxides such as ITO, nickel oxide, zinc oxide, cadmium tin oxide, indium oxide, tin oxide, magnesium oxide, ZnGa.sub.2O.sub.4, ZnO.sub.2/Sb, Ga.sub.2O.sub.3/Sn, AgInO.sub.2/Sn, In.sub.2O.sub.3/Zn, CuAlO.sub.2, LaCuOS, CuGaO.sub.2, and SrCu.sub.2O.sub.2. The choice of material used can depend on the location of the contacts and on the desired electrical characteristics, such as transparency, junction resistivity, and sheet resistance.

[0097] As described above, the LED chip **80** is arranged for flip-chip mounting. In operation, the p-contact **110** and n-contact **112** are bonded to a surface, such as a printed circuit board, with electrical paths for applying an electrical signal to the LED chip **80**. In most cases, the p-contact **110** and n-contact **112** are on the bottom surface, and light that is emitted toward the bottom of the LED chip risks being at least partially absorbed, such as by the printed circuit board. The first reflective layer **92** and the second reflective layer **94** are arranged below the active layer **88** so that light emitted toward the bottom is reflected back up to contribute to useful LED chip emission. The first reflective layer **92** reflects most light by TIR, with the majority of the remainder of the light being reflected by the second reflective layer **94**.

[0098] As described above, the actual thickness of the layers can be different, and variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are expected. A region illustrated or described as square or rectangular can have rounded or curved features, and regions shown as straight lines may have some irregularity. For example, FIG. **9** is a microscopic cross-sectional image of a portion of an LED chip according to some embodiments, such as FIG. **8**. In FIG. **9**, the active LED structure **82**, current-spreading layer **90**, first reflective layer **92**, hole **98**, second reflective layer **94**, barrier layer **102**, and passivation

layer **106** are visible. As shown in the image, the second reflective layer **94**, barrier layer **102**, and passivation layer **106** are on the first reflective layer **92** and are conformal to the first reflective layer **92** through the hole. The second reflective layer **94** partially fills the hole **98** with the remaining portion of the hole **98** filled by the barrier layer **102** and the passivation layer **106**. [0099] Accordingly, some embodiments disclosed herein include an LED chip comprising an active LED structure comprising an active layer between an n-type layer and a p-type layer; a first reflective layer adjacent the active LED structure and comprising a plurality of dielectric layers; a second reflective layer on the first reflective layer; a barrier layer on the second reflective layer; and a passivation layer on the barrier layer.

[0100] FIG. **10A** illustrates an LED chip **114** similar to the embodiments described above for FIG. **8**. In FIG. **10A**, the LED chip **114** is in a flip-chip orientation and includes an active LED structure **82**, a substrate **91**, a first reflective layer **92**, a second reflective layer **94**, a barrier layer **102**, a passivation layer **106**, vias **100**, active structure holes **104**, a p-contact **110**, and an n-contact **112** similar to those described in FIG. **8**. The substrate **91** is light transmissive (preferably transparent) and includes an outer major surface **118**, side surfaces **120**, and an internal surface **116**. The internal surface **116** is proximate the active LED structure **82** and includes a patterned surface **105** adjacent the active LED structure **82** having multiple recessed and/or raised features. In some embodiments, the patterned surface **105** is adjacent an n-layer of the active LED structure **82**. A patterned surface **105** is particularly useful in embodiments in which the substrate **91** comprises sapphire in order to promote extraction of light through the interface between the active LED structure **82** and the substrate **91**. In some embodiments, the passivation layer **106** includes a metal-containing interlayer **122** arranged therein, wherein the interlayer **122** may comprise Al or another suitable metal. Notably, the interlayer **122** is embedded within passivation layer **106** and is electrically isolated from the rest of the LED chip **114**. In application, the interlayer **122** may function as a crack stop layer for any cracks that may propagate through the passivation layer. Additionally, the interlayer **122** may reflect at least some light that may pass through both the first reflective layer **92** and the second reflective layer **94**.

[0101] FIG. **10B** illustrates the LED chip **114** of FIG. **10A** mounted to a submount **124** and covered with a layer of at least one lumiphoric material **126**. The submount **124** includes a first contact pad **128** and a second contact pad **130** arranged proximate to the p-contact **110** and n-contact **112** of the LED chip **114**, respectively. Solderless, soldered flux, direct attach, or other conventional attachment means may be used to establish conductive electrical communication between the first contact pad **128** and the p-contact **110** and the second contact pad **130** and the n-contact **112**. As illustrated in FIG. **10B**, the layer of at least one lumiphoric material **126** is arranged to cover the outer major surface **118**, side surfaces **120**, and at least a portion of the submount **124**.

[0102] Lumiphoric materials as described herein may be or include one or more of a phosphor, a scintillator, a lumiphoric ink, a quantum dot material, a day glow tape, and the like. Lumiphoric materials may be provided by any suitable means, for example, direct coating on one or more surfaces of an LED, dispersal in an encapsulant material configured to cover one or more LEDs, and/or coating on one or more optical or support elements (e.g., by powder coating, inkjet printing, or the like). In certain embodiments, lumiphoric materials may be downconverting or upconverting, and combinations of both downconverting and upconverting materials may be provided. In certain embodiments, multiple different (e.g., compositionally different) lumiphoric materials arranged to produce different peak wavelengths may be arranged to receive emissions from one or more LED chips. In some embodiments, one or more phosphors may include yellow phosphor (e.g., YAG:Ce), green phosphor (LuAg:Ce), and red phosphor (Ca.sub.i-x-ySr.sub.xEu.sub.yAlSiN.sub.3) and combinations thereof.

[0103] One or more lumiphoric materials may be provided on one or more portions of a flip-chip LED and/or a submount in various configurations. In certain embodiments, one or more surfaces of flip-chip LEDs may be conformally coated with one or more lumiphoric materials, while other

surfaces of such LEDs and/or associated submounts may be devoid of lumiphoric material. In certain embodiments, a top surface of a flip-chip LED may include lumiphoric material, while one or more side surfaces of a flip-chip LED may be devoid of lumiphoric material. In certain embodiments, all or substantially all outer surfaces of a flip-chip LED (e.g., other than contact-defining or mounting surfaces) are coated or otherwise covered with one or more lumiphoric materials. In certain embodiments, one or more lumiphoric materials may be arranged on or over one or more surfaces of a flip-chip LED in a substantially uniform manner; in other embodiments, one or more lumiphoric materials may be arranged on or over one or more surfaces of a flip-chip LED in a manner that is non-uniform with respect to one or more of material composition, concentration, and thickness. In certain embodiments, the loading percentage of one or more lumiphoric materials may be varied on or among one or more outer surfaces of a flip-chip LED. In certain embodiments, one or more lumiphoric materials may be patterned on portions of one or more surfaces of a flip-chip LED to include one or more stripes, dots, curves, or polygonal shapes. In certain embodiments, multiple lumiphoric materials may be arranged in different discrete regions or discrete layers on or over a flip-chip LED.

[0104] In certain embodiments, a lumiphoric material may be arranged over a light-transmissive surface of the substrate, and the substrate may comprise a thickness of at least 120 μm , at least 150 μm , at least 170 μm , at least 200 μm , at least 230 μm , at least 250 μm (with the preceding minimum thickness values optionally bounded at the upper end by any of the foregoing thickness values), or another thickness threshold specified herein. Without intending to be limited by any specific theory of operation, it is currently believed that providing a relatively thick substrate according to one or more of the preceding thickness thresholds may enhance conversion efficiency of a lumiphor-converted flip-chip LED due to one or more of the following phenomena: (i) increasing physical separation (distance) between an outer major (e.g., top) surface of a substrate and the multi-layer mirror, and (ii) reducing luminous flux on lumiphoric material arranged on or over an outer surface of the substrate.

[0105] Benefits of reduced optical losses may be more pronounced and significant with larger LED die sizes (e.g., with substrate widths of at least about 1.2 mm, at least about 1.4 mm, at least about 2 mm, or larger). Larger die have a greater dependency on internal reflectivity due to more interaction of light with the internal reflector layers before the light escapes the LED. Benefits of reduced optical losses may also be more pronounced in producing warm white (versus cool white) emitters, since warm white emitters typically involve a greater amount of back reflection into a LED chip (e.g., due to conversion by multiple phosphor materials or material layers, such as a yellow layer and a red layer). As described above, the first reflective layer **92** includes improved reflectivity across a wide wavelength range and across a wide angle of incidence range and is therefore additionally well suited for larger LED die and/or LED die with multiple lumiphors that convert light from the active LED structure **82** to multiple other colors. Additionally, the first reflective layer **92** in combination with any or all of the second reflective layer **94**, the interlayer **122**, the patterned surface **105**, and the light transmissive or transparent substrate **91** reduces optical losses within the LED chip **114** providing an increase in brightness or luminous flux.

[0106] FIG. **11A** illustrates an LED chip **132** according other embodiments. In FIG. **11A**, an active LED structure **82**, a substrate **91**, a first reflective layer **92**, a second reflective layer **94**, a barrier layer **102**, a passivation layer **106**, vias **100**, active structure holes **104**, a p-contact **110**, and an n-contact **112** are similar to those described in FIG. **10A**. Additionally, the first reflective layer **92** and the second reflective layer **94** include portions that are proximate to the active LED structure **82** and “wraparound” peripheral portions of the the active LED structure **82** (including the n-type layer, active layer, and p-type layer). As shown in FIG. **11A**, the active LED structure **82** extends away from the substrate **91** and forms a mesa **134** with a mesa sidewall **134'** that is laterally bounded by at least one recess **136** at the periphery of the LED chip **132**. The at least one recess **136** includes a peripheral wraparound portion **92'** of the first reflective layer **92** that bounds

peripheral portions of the active LED structure **82** forming the mesa **134**. Additionally, the at least one recess **136** includes a peripheral wraparound portion **94'** of the second reflective layer **94** that is arranged in contact with a portion of a wraparound portion **92'** of the first reflective layer **92**. Within the at least one recess **136**, the wraparound portion **92'** and peripheral wraparound portion **94'** are peripherally bounded by passivation material of the passivation layer **106**. Accordingly, the active LED structure **82** comprise a mesa sidewall **134'** and the first reflective layer **92** extends along the mesa sidewall **134'**.

[0107] As with previous embodiments, the first reflective layer **92** may comprise a plurality of first and second dielectric layers with different materials, such as silicon dioxide and silicon nitride, and the wraparound portion **92'** would therefore also comprise silicon dioxide and silicon nitride. Accordingly, the peripheral portions of the LED active structure **82** have improved reflectivity and more light may be redirected toward the substrate **91** and out of the LED chip **132**. The presence of silicon nitride in the wraparound portion **92'** of the first reflective layer **92** serves to provide passivation to the side of the mesa **134** of active LED structure **82**. In particular, the presence of silicon nitride in the wraparound portion **92'** may serve to protect the active LED structure **82** from migration of metals, such as silver, from the second reflective layer **94**. Any metal migrating along the mesa **134** edge may contact both the n-layer and the p-layer of the active LED structure **82** and provide an electrical short that would cause failure of operation of the LED chip **132**. Additionally, the presence of silicon nitride in the passivation layer **106** within the recess **136** may serve to block potential paths for moisture to be drawn into contact with metal-containing portions of the second reflective layer **94**, which would be expected to lead to detrimental chemical interaction. Accordingly, the LED chip **132** is expected to have improved lumen maintenance, or less light loss over time, in all operating conditions.

[0108] FIG. **11B** illustrates the LED chip **132** of FIG. **11A** mounted to a submount **124** and covered with a layer of at least one lumiphoric material **126** similar to that of FIG. **10B**. The submount **124** includes a first contact pad **128** and a second contact pad **130** arranged proximate to the p-contact **110** and n-contact **112**, respectively. Solderless, soldered flux, direct attach, or other conventional attachment means may be used to establish conductive electrical communication between the first contact pad **128** and the p-contact **110** and the second contact pad **130** and the n-contact **112**. As illustrated in FIG. **11B**, the layer of at least one lumiphoric material **126** is arranged to cover an outer major surface **118** of the substrate **91** and one or more side surfaces **120** of the substrate **91** as well as the at least a portion of the submount **124**.

[0109] FIG. **12A** illustrates another embodiment of a LED chip **138** according to the present invention that is flip-chip mounted onto a submount or substrate for use. The LED chip has many layers similar to those in the embodiment shown in FIG. **8**, FIG. **10A**, and FIG. **10B** and described above, including an active LED structure **140** comprising an p-type layer **142**, n-type layer **144**, and an active layer **146**. A current-spreading layer **148** is included on the p-type layer **142** to spread current to the p-type layer **142** during operation. A first reflective layer **150** is included on the current-spreading layer **148**, and a second reflective layer **152** is included on the first reflective layer **150** with an adhesion layer **154** between the two.

[0110] The first reflective layer **150** is arranged on the current-spreading layer **148** and adjacent the p-type layer **142** and can have any of the embodiments with a plurality of layers previously described for the first reflective layer **56** as described for FIG. **5B** or the first reflective layer **92** as described for FIG. **8**. For example, in LED chip **138**, the first reflective layer **150** may comprise a plurality of dielectric layers. The plurality of dielectric layers may comprise embodiments in which each dielectric layer of the plurality of dielectric layers comprises a different thickness. In some embodiments, the first reflective layer **150** comprises from 7 to 13 dielectric layers. In some embodiments, the first reflective layer **150** comprises a plurality of alternating first dielectric layers and second dielectric layers. The plurality first dielectric layers may have a different index of refraction than the plurality of second dielectric layers. In further embodiments, the plurality of first

dielectric layers comprises silicon dioxide and the plurality of second dielectric layers comprises silicon nitride, although other material combinations are possible as described above. In some embodiments, the first reflective layer **150** comprises an uneven number of first dielectric layers and second dielectric layers, while in other embodiments, the first reflective layer **150** comprises an even number of first dielectric layers and of second dielectric layers. In some embodiments, the first reflective layer **150** comprises an aperiodic Bragg reflector. In some embodiments, the first reflective layer **150** comprises a plurality of dielectric layers in which the thickest dielectric layer of the plurality of dielectric layers is spaced from the active layer by at least one thinner dielectric layer. In some embodiments, the first reflective layer **150** comprises a plurality of first dielectric layers having an average thickness that is greater than an average thickness of a plurality of second dielectric layers, and at least one layer of the plurality of second dielectric layers has a thickness greater than at least one layer of the plurality of first dielectric layers. All other embodiments described for the first reflective layer **56** of FIG. **5** and the first reflective layer **92** of FIG. **8** are also applicable to the first reflective layer **150** of FIG. **12A**.

[0111] The LED chip **138** further comprises reflective layer holes **156** that can pass through the adhesion layer **154** and the first reflective layer **150** to the current-spreading layer **148**. The reflective layer holes **156** can then be filled or partially filled when the second reflective layer **152** is deposited. Accordingly, the second reflective layer **152** is formed on the first reflective layer **150** and comprises vias **158** to the current-spreading layer **148** as previously described for FIG. **8**.

[0112] In FIG. **12A**, a passivation layer **160** and a barrier layer **162** extend beyond the edge of the active LED structure **140** where a p-contact **164** can be formed on the barrier layer **162**. It is understood that the passivation layer **160** may include a metal interlayer as described in FIG. **10A**. An active structure hole **166** is included through the adhesion layer **154**, the first reflective layer **150**, the current-spreading layer **148**, the p-type layer **142**, and the active layer **146**. The passivation layer **160** is included on the barrier layer **162** and the side surfaces of the active structure hole **166**, and an n-contact via or n-contact **168** is included in the active structure hole **166** for applying an electrical signal to the n-type layer **144**. An electrical signal applied to the p-contact **164** is conducted to the p-type layer **142** through the barrier layer **162**, the second reflective layer **152**, and the current-spreading layer **148**. Accordingly, an electrical signal applied across the p-contact **164** and the n-contact **168** is conducted to the p-type layer **142** and the n-type layer **144**, causing the active layer **146** to emit light.

[0113] In FIG. **12A**, the growth substrate for LED chip **138** has been removed, and the top surface **170** of the n-type layer **144** is textured for light extraction. To provide mechanical stabilization, the LED chip **138** is flip-chip mounted to a submount **172**, with a bond metal layer **174** and blanket mirror layer **176** between the submount **172** and the active LED structure **140**. Accordingly, the p-type layer **142** is between the submount **172** and the active layer **146**. The submount **172** can be made of many different materials, with a suitable material being silicon. The blanket mirror layer **176** can be made of many different materials, with a suitable material being Al. The blanket mirror layer **176** helps to reflect LED light that escapes reflection by the first reflective layer **150** and the second reflective layer **152**, such as light that may pass through the active structure hole **166**. In some embodiments, the first reflective layer **150** may wrap around and extend on the side of the active LED structure **140** within the active structure hole **166**.

[0114] FIG. **12B** illustrates the LED chip **138** of FIG. **12A** covered with a layer of at least one lumiphoric material **178**. The lumiphoric material **178** may be any material or combination of materials as described for the layer of at least one lumiphoric material **126** of FIG. **10B**. In FIG. **12B**, the layer of at least one lumiphoric material **178** may be deposited on top of active LED structure **140** while leaving the p-contact **164** exposed. As with the embodiments of FIG. **12A**, the active LED structure **140** includes an n-type layer **144**, an active layer **146**, and a p-type layer **142**. In some embodiments, the layer of at least one lumiphoric material **178** is on the n-type layer **144**. As described above, the first reflective layer **92** includes improved reflectivity across a wide

wavelength range and across a wide angle of incidence range. In operation, light of various wavelengths is emitted omnidirectionally from the active layer **146** and/or converted omnidirectionally by the layer of at least one lumiphoric material **178** and may be reflected by the first reflective layer **150**, the second reflective layer **152** or the blanket mirror layer **176** and extracted from the textured top surface **170**. Accordingly, the first reflective layer **150** in combination with any or all of the second reflective layer **152**, the blanket mirror layer **176**, and the textured top surface **170** reduces optical losses within the LED chip **138**, providing an increase in brightness or luminous flux.

[0115] In addition to the LED chip embodiments previously described, reflective layers described herein may also provide reflectivity improvements in other configurations. For example, FIG. **13** is a cross-sectional representation of a packaged LED **180** according to some embodiments. In FIG. **13**, at least one light source **182**, such as an LED chip, mounted on a submount **184**. The submount **184** may include any number of materials, including but not limited to, alumina, AlN, silicon, and printed circuit boards. In some embodiments, a first reflective layer **186** is on the submount **184** and between the at least one light source **182** and the submount **184**. In further embodiments, the first reflective layer **186** extends on the submount **184** beyond where the at least light source **182** is mounted. In other embodiments, the first reflective layer **186** may only be on portions of the submount **184** in areas outside of where the at least one light source **182** is mounted. The packaged LED **180** may further include a lumiphoric layer **188** and an encapsulant **190**. The lumiphoric layer **188** may include any of the lumiphoric materials previously described, for example as described for FIG. **10A**, and the encapsulant **190** may include an optically transmissive material such as silicone or glass that may be molded in the shape of a lens. In some embodiments, the lumiphoric layer **188** is on the first reflective layer **186** outside of where the at least one light source **182** is mounted. In some embodiments, the lumiphoric layer **188** and the encapsulant **190** may be combined, for example a silicone material acting as a binder for lumiphoric materials. The first reflective layer **186** may be any of the multiple layer reflective combinations as previously described, for example as described for the first reflective layer **56** of FIG. **5B**, FIG. **5C**, and FIG. **5D**. Accordingly, the first reflective layer **186** is in an optical path of the at least one light source **182**. For example, light emitted by the at least one light source **182** and light converted by the lumiphoric layer **188** toward the submount **184** may be reflected by the first reflective layer **186** in locations between the at least one light source **182** and the submount **184** as well as locations on the submount **184** outside of where the at least light source **182** is mounted.

[0116] FIG. **14** is a cross-sectional representation of a packaged LED **192** according to some embodiments that is similar to the packaged LED **180** of FIG. **13**, but is illustrated with a plurality of light sources **194**, such as LED chips. Notably, a first reflective layer **195** is on a submount **196** and is in an optical path of the plurality of light sources **194** and is configured to reflect light emitted by the plurality of light sources **194**. The first reflective layer **195** may be any of the multiple layer reflective combinations as previously described, for example as described for the first reflective layer **56** of FIG. **5B**, FIG. **5C**, and FIG. **5D**. An encapsulant **198** may be provided over the plurality of LED chips **194** and may comprise many shapes, for example but not limited to, a square or rectangular cubic shape, a hemispherical-shaped lens, or a hemispherical-shaped lens with planar side surfaces. In some embodiments, the encapsulant may be dispensed on the submount **196** inside of a retention material (not shown) that surrounds the plurality of light sources **194**.

[0117] FIG. **15** is a cross-sectional representation of a multiple-junction LED chip **200** according to some embodiments. The multiple-junction LED chip **200** includes a substrate **204**, at least one n-type layer **206**, at least one active layer **208**, and at least one p-type layer **210**. Individual junctions **218** are provided by isolation trenches **216** that extend through the at least one p-type layer **210** and the at least one active layer **208** to the at least one n-type layer **206**. In some embodiments, the isolation trenches **216** extend to the substrate **204**. In some embodiments, the isolation trenches **216**

extend completely through the substrate **204** as illustrated by the vertical dashed lines in FIG. **15**. In embodiments where the substrate **204** is removed, the isolation trenches **216** may extend through the at least one n-type layer **206**. Individual junctions **218** are individually addressable by way of a separate first contact **212** and a separate second contact **214** for each of the individual junctions **218**. For example, the multiple-junction LED chip **200** may be mounted on a submount (not shown) that includes corresponding electrical connections for the first contact **212** and the second contact **214**. The submount may be a printed circuit board with electrical traces or any other type of submount with corresponding electrical connections. The multiple-junction LED chip **200** further includes a first reflective layer **202** that may be on the at least one p-type layer **210** of the individual junctions **218**. The first reflective layer **202** may be any of the multiple layer reflective combinations as previously described, for example as described for the first reflective layer **56** of FIG. **5B**, FIG. **5C**, and FIG. **5D**. The first reflective layer **202** is in an optical path of the multiple-junction LED chip **200**. The multiple-junction LED chip **200** may be a flip-chip LED similar to the embodiments previously described for FIG. **8**, FIG. **10A**, FIG. **10B**, FIG. **11A**, and FIG. **11B**. In other embodiments, the multiple-junction LED chip **200** may be similar to the embodiment previously described for FIG. **12A** and FIG. **12B**. The first reflective layer **202** may be on each of the individual junctions **218** and the isolation trenches **216** may extend through the first reflective layer. In other embodiments, the first reflective layer **202** may be a continuous layer across all of the individual junctions **218** as shown by the horizontal dashed lines in FIG. **15**. For example, the isolation trenches **216** may be formed from the substrate or the at least one n-type layer **206** toward but not through the first reflective layer **202**.

[0118] In other embodiments, reflective layers described herein may also provide reflectivity improvements in system level configurations. For example, FIG. **16** is a cross-sectional representation of a portion of a lighting fixture **220** according to some embodiments. In FIG. **16**, a lighting fixture **220** includes a light source **222**. The light source **222** may be a single light source or a plurality of light sources and may include a packaged LED, a semiconductor-based LED chip, an organic LED chip, a laser, a fluorescent light source, and an incandescent light source among others. The light source **222** is mounted or supported to a housing **224** of the light fixture **220**. A first reflective layer **226** is on the housing and may be located between the light source **222** and the housing **224** as well as in locations outside of where the light source **222** is supported by the housing **224**. In some embodiments, the first reflective layer **226** is only in locations outside of where the light source **222** is supported by the housing **224**. The first reflective layer **226** may be any of the multiple layer reflective combinations as previously described, for example as described for the first reflective layer **56** of FIG. **5B**, FIG. **50**, and FIG. **5D**. The lighting fixture **220** may further include a light-transmissive cover **228**. Accordingly, the first reflective layer **226** is in an optical path of the light source **222**. For example, light emitted by the light source **222** may be reflected by the first reflective layer **226** on the housing **224** before exiting the lighting fixture **220** through the light-transmissive cover **228**.

Accordingly, some embodiments described herein include a device comprising a light source and a first reflective layer in an optical path of the light source. The first reflective layer comprises a plurality of first dielectric layers and a plurality of second dielectric layers and an average thickness of the plurality of first dielectric layers is greater than an average thickness of the plurality second dielectric layers. At least one layer of the plurality of second dielectric layers comprises a thickness greater than at least one layer of the plurality of first dielectric layers. Additionally, the reflective layer may comprise any of the embodiments described above for FIG. **5B**, FIG. **5C**, and FIG. **5D**. Other embodiments described herein include a device comprising a light source and a first reflective layer in an optical path of the light source. The first reflective layer comprises a plurality of first dielectric layers and a plurality of second dielectric layers wherein each layer of the plurality of first dielectric layers comprises a different thickness. A thickest layer of the plurality of first dielectric layers is between other layers of the plurality of first dielectric layers.

Additionally, the reflective layer may comprise any of the embodiments described above for FIG. 5B, FIG. 5C, and FIG. 5D.

[0119] Those skilled in the art will recognize improvements and modifications to the preferred embodiments of the present disclosure. All such improvements and modifications are considered within the scope of the concepts disclosed herein and the claims that follow.

Claims

1. A light-emitting diode (LED) chip comprising: an active LED structure comprising an active layer between an n-type layer and a p-type layer; and a first reflective layer adjacent the active LED structure and comprising a plurality of first dielectric layers and a plurality of second dielectric layers, wherein each layer of the plurality of first dielectric layers comprises a different thickness; wherein an average thickness of the plurality of first dielectric layers is greater than an average thickness of the plurality second dielectric layers; wherein at least one layer of the plurality of second dielectric layers comprises a thickness greater than at least one layer of the plurality of first dielectric layers; and wherein an average reflectivity of the first reflective layer for a wavelength of 460 nanometers is above 93% for angles of incidence in a range from 15 degrees to 50 degrees.
2. The LED chip of claim 1, wherein the average thickness of the plurality of first dielectric layers is greater than the average thickness of the plurality second dielectric layers by at least a factor of 2.
3. The LED chip of claim 1, wherein the average thickness of the plurality of first dielectric layers is greater than the average thickness of the plurality second dielectric layers by at least a factor of 3.
4. The LED chip of claim 1, wherein a thickest layer of the plurality of first dielectric layers is closer to the active LED structure than any other layer of the plurality of first dielectric layers.
5. The LED chip of claim 4, wherein the thickest layer of the plurality of first dielectric layers is at least five times thicker than the thinnest layer of the plurality of first dielectric layers.
6. The LED chip of claim 1, wherein a thickest layer of the plurality of first dielectric layers is between other layers of the plurality of first dielectric layers.
7. The LED chip of claim 6, wherein the thickest layer of the plurality of first dielectric layers is at least five times thicker than the thinnest layer of the plurality of first dielectric layers.
8. The LED chip of claim 1, wherein the plurality of first dielectric layers comprises at least one of an oxide of silicon, an oxide of titanium, and an oxide of tantalum.
9. The LED chip of claim 1, further comprising a second reflective layer on the first reflective layer, wherein the second reflective layer forms an electrical path through the first reflective layer.
10. The LED chip of claim 9, further comprising a current-spreading layer between the active LED structure and the first reflective layer, wherein the second reflective layer forms the electrical path through the first reflective layer to electrically connect with the current-spreading layer.
11. A light-emitting diode (LED) chip comprising: an active LED structure comprising an active layer between an n-type layer and a p-type layer; and a first reflective layer adjacent the active LED structure and comprising a plurality of first dielectric layers and a plurality of second dielectric layers, wherein each layer of the plurality of first dielectric layers comprises a different thickness; wherein an average thickness of the plurality of first dielectric layers is greater than an average thickness of the plurality second dielectric layers; wherein at least one second dielectric layer of the plurality of second dielectric layers is at least two times thicker than at least one first dielectric layer of the plurality of first dielectric layers; and wherein an average reflectivity of the first reflective layer for a wavelength of 460 nanometers is above 93% for angles of incidence in a range from 15 degrees to 50 degrees.
12. The LED chip of claim 11, further comprising a second reflective layer on the first reflective

- layer, wherein the second reflective layer forms an electrical path through the first reflective layer.
- 13.** The LED chip of claim 12, further comprising an adhesion layer between the first reflective layer and the second reflective layer.
- 14.** The LED chip of claim 11, wherein the average thickness of the plurality of first dielectric layers is greater than the average thickness of the plurality second dielectric layers by at least a factor of 2.
- 15.** The LED chip of claim 11, wherein the thickest layer of the plurality of first dielectric layers is at least five times thicker than the thinnest layer of the plurality of first dielectric layers.
- 16.** A light-emitting diode (LED) chip comprising: an active LED structure comprising an active layer between an n-type layer and a p-type layer; and a first reflective layer adjacent the active LED structure and comprising a plurality of first dielectric layers that alternate with a plurality of second dielectric layers wherein each layer of the plurality of first dielectric layers comprises a different thickness; wherein a thickest layer of the plurality of first dielectric layers is at least four times thicker than any other first dielectric layer of the plurality of first dielectric layers; and wherein an average reflectivity of the first reflective layer for a wavelength of 460 nanometers is above 93% for angles of incidence in a range from 15 degrees to 50 degrees.
- 17.** The LED chip of claim 16, further comprising a second reflective layer on the first reflective layer, wherein the second reflective layer forms an electrical path through the first reflective layer.
- 18.** The LED chip of claim 17, further comprising a current-spreading layer between the active LED structure and the first reflective layer, wherein the second reflective layer forms the electrical path through the first reflective layer to electrically connect with the current-spreading layer.
- 19.** The LED chip of claim 17, further comprising an adhesion layer between the first reflective layer and the second reflective layer.
- 20.** The LED chip of claim 16, wherein an average thickness of the plurality of first dielectric layers is greater than an average thickness of the plurality second dielectric layers, and at least one layer of the plurality of second dielectric layers comprises a thickness greater than at least one layer of the plurality of first dielectric layers.
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